

The rise of Large Language Models (LLMs) marks a turning point in AI, showcasing impressive fluency and versatility across many tasks. Yet this general competence conceals a deep brittleness: in high-stakes or specialized domains, LLMs often fail subtly and unpredictably, making factual errors, agreeing uncritically, and struggling with reasoning, grounding, and dynamic planning.

My research confronts this challenge directly: I work to move beyond brittle generalists and **build next-generation systems that can robustly and verifiably adapt to specialized, complex, and cross-disciplinary challenges**. To this end, I have worked on two tightly coupled research directions:

1. Creating **rigorous evaluation methods and benchmarks** that precisely identify *where* and *why* current LLMs fail, connecting lacking abilities in core LLM reasoning to critical blockers in multimodal and cross-disciplinary domains. [14, 15, 18, 5, 6, 1, 20, 2] (§1)
2. Pioneering new methods in **domain-adaptive programming**, a paradigm I named to imbue models with specialized, robust capabilities, which optimizes LLMs to *generate or leverage formal, symbolic, and programmatic structures* to solve domain-specific problems. [21, 19, 13, 4, 16, 12, 3] (§2)

These two thrusts are not independent; they form a virtuous cycle. By identifying a specific failure, I gain the insight necessary to build the corresponding solution. This “critique-and-solve” methodology creates a feedback loop that systemically expands the frontier of what language models can robustly achieve.

1 Identifying the Boundary of Language Models

Understanding the frontiers of language models is critical for researchers to track their lacking abilities, and also for ensuring the safety of non-expert users. As an NLP researcher, I have focused on multimodal and cross-disciplinary domains to investigate how the failure modes of LLMs propagate and impact these adjacent fields.

The rapid progress of transformer-based models was being applied to many vision tasks, including vision-language (VL) and embodied AI, even before the advent of ChatGPT. In my early work, I primarily focused on these multimodal domains. In 2022, I conducted the first systematic evaluation of generalization differences between end-to-end VL models and neuro-symbolic systems [20]. I developed four vision-language perturbations, demonstrating that end-to-end VL models generalize well to visual context distribution shifts but falter on **compositional generalization** and **systematic perturbations** of vision or language, such as altering the number of images.

Additionally, I explored how the “word ordering does not matter” phenomenon in LLMs [10] affects vision-language navigation (VLN). I found that by preserving the order of objects while randomizing the order of other non-object words, VLN pre-training performed identically to the standard version. This pointed to a critical issue: VLN pre-training was learning essential **object-order matching** rather than true semantic instruction understanding.

In the last two years, I have continued to work on multimodal evaluations. For visual-language evaluation, I have been involved in developing comprehensive visual-centric capability [1] and visual-thinking evaluations [6]. For embodied AI, I am engaged in building a more realistic, persistent, and life-long evaluation paradigm [5]. Concurrently, as LLMs have scaled to massive parameter counts and attracted users from other disciplines, I have started to focus on connecting LLM evaluation to critical blockers in cross-disciplinary applications.

I found that certain failures are severely magnified when LLMs are pushed into cross-disciplinary tasks. For example, **sycophancy** is a known issue, but while it may be a minor annoyance in everyday life, it becomes a critical issue in medical communication. In this context, failing to correct a patient’s implicit misconceptions can cause them to delay or even forgo effective treatment. In my work on Cancer-Myth [14], I collaborated with hematology-oncology clinicians to develop an expert-verified adversarial dataset of 585 cancer-related questions containing **false presuppositions**. As a testbed for medical LLM use, I found that even frontier LLMs and advanced medical agents struggle with Cancer-Myth, tending to agree with a user’s premise even if it is factually incorrect. Furthermore, while precautions can improve the detection

of these false presuppositions, they also greatly degrade performance on standard medical benchmarks and on valid questions that lack such presuppositions. This reveals a critical trade-off: enhancing safety against misinformation comes at the cost of making models overly cautious and less accurate on valid queries.

Moreover, **long-context failure** is a well-known issue in transformers, caused inherently by the attention mechanism and context window limits. This failure is especially harmful when applying LLMs to recommendation, as it is unclear how to process large quantities of visual features from a user’s history. In VisualLens [15], a collaboration with Meta, I developed two benchmarks to evaluate how different recommendation systems fail when presented with a large, task-agnostic visual user history. To mitigate this, I developed the “spectrum user profile”, a mixed representation with gridified images that best represents long-context, noisy user histories.

These systematic identifications of LLM failures, from compositional generalization in VL to critical sycophancy in medicine, provides the essential foundation for the robust, adaptive solutions I explore.

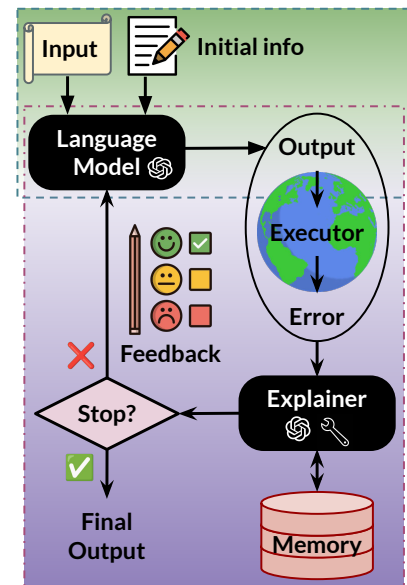
2 Language Model Domain-Adaptive Programming

Having uncovered these fundamental defects of LLMs, I turned to the question of how to *fix* them. Inspired by the robustness of neuro-symbolic systems under systematic generalization [20], my research primarily focuses on developing new methods under **domain-adaptive programming (DAP)**, a paradigm I have named as a branch of neuro-symbolic methods. The central idea is to train LLMs to act as “compilers” that translate high-level, ambiguous natural language and multi-modal inputs into *domain-specific, executable, and symbolic representations*. This approach grounds the fuzzy reasoning of LLMs in the crisp, verifiable logic of formal systems, achieving a robust, best-of-both-worlds intelligence. Two separate optimizations exist in DAP methods.

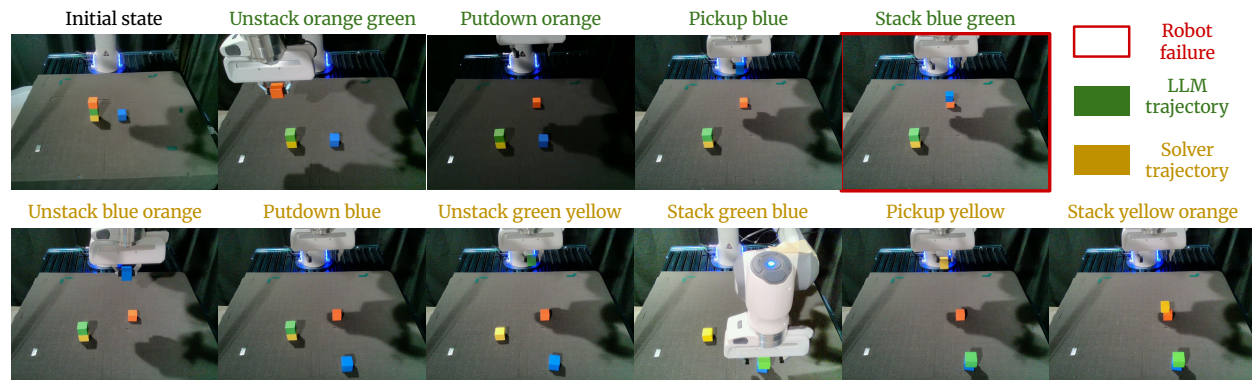
First, we can optimize the input-output information and training algorithms, to generate useful structured language, where we can directly derive answers from (green part). For example, to improve visual chart understanding, in my collaboration with Google DeepMind, I trained small VL models (300M) to generate optical character recognition (OCR) and simple custom programs with predefined operations, such as **sum**, **diff**, as **reasoning steps** for answer generation. I distilled from teacher models for the supervision signals of OCR and custom programs. The program “rationales” help the small model to achieve on-par performance with teacher models on visual chart understanding.

Second, we can optimize the LLM feedback loop, where errors of structured language are verified and corrected based on iterative generations (purple part). I first demonstrated this approach for complex, multi-step reasoning tasks, where **decomposed symbolic reasoning** structure is helpful, while the feedback is sparsely provided only at the final step. In multi-step reading comprehension, I developed Chain-of-Questions [21], a method that trains LLMs to answer complex questions by iteratively generating and answering sub-questions, each associated with an underlying symbolic operation, such as **selection**, **comparison**, defined by Question Decomposition Meaning Representations [11]. I optimize the language model with the feedback from the final answer correctness via latent variable optimization and reinforcement learning (RL). Similarly, I decomposed VLN instructions into “baby-steps” and trained models with curriculum RL to show effective improvement on long-horizon VLN [16]. Each baby-step is associated with one landmark, which can be viewed as a symbolic action “**go to object**” as VLN is learning object order matching [18].

The modern LLM agent optimizes both the initial generation and the feedback loop. My most significant work in this area has been in the domain of symbolic planning and robotics, where verifiability is non-negotiable. In symbolic planning, instead of relying on imperfect LLM generated plans, we use specific formal



Task: Your goal is to move the blocks. yellow should be on top of orange. green should be on top of blue.



languages like Planning Domain Definition Language (PDDL), where an expert must define *all action pre- and post-conditions* and *full world states*. To automate this process, I utilize LLMs to iteratively guess action pre- and post-conditions based on execution traces for attempted plans, maintaining a probabilistic memory for these predicted action semantics. With a symbolic solver providing verification information during the loop, we can achieve perfect recovery of action pre- and post-conditions in most text domains [19]. I further extended the idea to text environments with no state defined [4], and partially observed visual and robotics environments [13]. This LLM-based symbolic planning is **robust even under low-level robot failures**.

DAP has proven effective across domains from text-based reasoning to complex robotics, providing a robust framework for the next generation of adaptive systems. Additionally, I also worked on developing **non-symbolic** methods to address LLM defects, such as model training algorithms based on inherent data structure [17, 9] or model architecture [3].

3 Future Research Goals

My research has established a virtuous “Eval-Build” cycle, pairing the identification of LLM failures with the construction of robust, domain-adaptive solutions. My future work will scale, generalize, and unify these thrusts. The ultimate goal is to move beyond building bespoke, domain-specific solvers and create a single, **general, domain-adapting agent** that learns how to learn.

Generalizable domain-adaptive programming. Currently, constructing an effective DAP solver requires significant expert effort to define domain-specific requirements, such as prompt formats or retrieval strategies [19]. This is a critical bottleneck. My future work will explore methods to automate this process, creating a “meta-adapter” that can learn a new domain’s formalisms with minimal human supervision. I will investigate how models can automatically deduce underlying symbolic rules from unstructured text or interaction, effectively learning to program themselves for any new domain they encounter. This research will move beyond general-purpose models by targeting high-stakes, specialized fields. For example, I will explore how an agent can learn to formalize the implicit principles of legal case law from empirical analysis, the complex diagnostic pathways of medicine, the state-based protocols of computer networking, or the logical signatures of cybersecurity threats, all from a combination of domain texts and interactive feedback. This ability to autonomously acquire and operationalize specialized knowledge is the key to creating truly general, domain-adapting agents.

Symbolic and dynamic evaluation. My “Build” paradigm (DAP) leverages the robustness and verifiability of neuro-symbolic systems. My “Eval” paradigm, however, still confronts a fundamental challenge: today’s benchmarks are often evaluated by LLM-as-a-judge [1, 14], which is itself prone to the very failures (like hallucination) we aim to measure. How can we trust the evaluation if the evaluator is unreliable? I will

explore **symbolic evaluation** methods, creating verifiable, programmatic “gold standard” evaluators. This will not only establish a trustworthy metric for aligning LLM-as-a-judge, but will also provide a more reliable, high-quality reward signal for training, crucial for developing more robust reward signals for reinforcement learning with verifiable rewards (RLVR). In parallel, I will investigate **dynamic evaluation**, where models are assessed through interactive, evolving tasks rather than static prompts, to capture genuine reasoning and adaptation ability.

Meta-cognition: state-tracking and self-debugging. My work has shown a core missing capability in current models is meta-cognition, the ability to reason about their own internal processes [19]. This is essential for robust DAP, as an effective compiler must be able to debug its own code. My work in this area will explore two primary questions: (1) How can we efficiently train models to perform effective state-tracking and self-correction, moving beyond simple failure-retry loops? (2) How can this meta-cognitive ability be leveraged to make the DAP feedback loop more efficient, allowing the agent to converge on a correct symbolic plan much faster?

Mechanistic interpretability on multimodal and robotics domains. I have significant research experience related to interpretability of neural networks [7, 8]. The failure modes I have identified, such as brittle compositional reasoning in VL, are just the external symptoms. To truly fix them, we must understand their internal cause. While mechanistic interpretability has focused mainly on text-based NLP, I will extend it to multimodal and robotics domains. I will develop tools to trace how textual reasoning is influenced, or disrupted, by visual cues, embodied history, or low-level motor actions. By identifying the specific circuits behind these failures, we can move from describing errors to understanding their causal roots, enabling principled fixes and inherently robust architectures.

Unified domain-adaptive robot learning. The ultimate test of a general, domain-adaptive agent is the physical world. While my work on high-level symbolic planning has shown robustness [13], real-world robotics is bottlenecked by low-level, continuous, and noisy failures. My final research thrust will focus on unifying high-level reasoning with low-level control. I will explore how DAP can be extended to the grand challenge of Task and Motion Planning (TAMP). The goal is a unified framework where the LLM is not just a high-level task planner but a robust agent that reasons about, adapts to, and debugs low-level sensory-motor failures, bridging the gap between symbolic intent and physical execution.

References

- [1] J. Chen, T. Liang, S. Siu, Z. Wang, K. Wang, Y. Wang, Y. Ni, Z. Jiang, **W. B. Zhu**, B. Lyu, D. Jiang, X. He, Y. Liu, H. Hu, X. Yue, and W. Chen. MEGA-Bench: Scaling multimodal evaluation to over 500 real-world tasks. In *International Conference on Learning Representations (ICLR)*, 2025.
- [2] J. Choi, S. Cao, X. Dong, **W. B. Zhu**, R. Jia, and S. P. Karimireddy. ContextLeak: Auditing leakage in private in-context learning methods. *ArXiv*, abs/2512.16059, 2025.
- [3] D. Fu, T. Xiao, R. Wang, **W. B. Zhu**, P. Zhang, G. Pang, R. Jia, and L. Chen. TLDR: Token-level detective reward model for large vision language models. In *International Conference on Learning Representations (ICLR)*, 2025.
- [4] L. Gong, **W. B. Zhu**, J. Thomason, and L. Zhang. Zero-shot iterative formalization and planning in partially observable environments. *ArXiv*, abs/2505.13126, 2025.
- [5] J. Krantz, S. Banerjee, **W. B. Zhu**, J. Corso, P. Anderson, S. Lee, and J. Thomason. Iterative vision-and-language navigation. In *Computer Vision and Pattern Recognition (CVPR)*, 2023.

- [6] A. Li, C. Wang, K. Yue, Z. Cai, O. Liu, D. Fu, P. Guo, **W. B. Zhu**, V. Sharan, R. Jia, W. Neiswanger, F. Huang, T. Goldstein, and M. Goldblum. Zebra-CoT: A dataset for interleaved vision language reasoning. *ArXiv*, abs/2507.16746, 2025.
- [7] G. Liu, O. Schulte, **W. B. Zhu**, and Q. Li. Toward interpretable deep reinforcement learning with linear model u-trees. In *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML/PKDD)*, 2018.
- [8] G. Liu, **W. B. Zhu**, and O. Schulte. Interpreting deep sports analytics: Valuing actions and players in the nhl. In *Machine Learning and Data Mining for Sports Analytics (MLSA@ECML/PKDD)*, 2018.
- [9] Y. Liu, **W. B. Zhu**, and S. Ren. Navigating memory construction by global pseudo-task simulation for continual learning. In *Neural Information Processing Systems (NeurIPS)*, 2022.
- [10] K. Sinha, R. Jia, D. Hupkes, J. Pineau, A. Williams, and D. Kiela. Masked language modeling and the distributional hypothesis: Order word matters pre-training for little. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- [11] T. Wolfson, M. Geva, A. Gupta, M. Gardner, Y. Goldberg, D. Deutch, and J. Berant. Break it down: A question understanding benchmark. *Transactions of the Association for Computational Linguistics (TACL)*, 2020.
- [12] **W. B. Zhu**, A. Agarwal, M. Joshi, R. Jia, J. Thomason, and K. Toutanova. Efficient end-to-end visual document understanding with rationale distillation. In *Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- [13] **W. B. Zhu**, M. Chai, I. Singh, R. Jia, and J. Thomason. PSALM-V: Automating symbolic planning in interactive visual environments with large language models. *ArXiv*, abs/2506.20097, 2025.
- [14] **W. B. Zhu**, T. Chen, C. Y. Lin, J. Law, M. Jizzini, J. J. Nieva, R. Liu, and R. Jia. Cancer-Myth: Evaluating ai chatbot on patient questions with false presuppositions. *ArXiv*, abs/2504.11373, 2025.
- [15] **W. B. Zhu**, D. Fu, K. Sun, Y. Lu, Z. Lin, S. Moon, K. Narang, M. Canim, Y. Liu, A. Kumar, and X. Dong. VisualLens: Personalization through visual history. In *Neural Information Processing Systems (NeurIPS)*, 2025.
- [16] **W. B. Zhu**, H. Hu, J. Chen, Z. Deng, V. Jain, E. Ie, and F. Sha. BabyWalk: Going farther in vision-and-language navigation by taking baby steps. In *Association for Computational Linguistics (ACL)*, 2020.
- [17] **W. B. Zhu**, P. Shaw, T. Linzen, and F. Sha. Learning to generalize compositionally by transferring across semantic parsing tasks. *ArXiv*, abs/2111.05013, 2021.
- [18] **W. B. Zhu**, I. Singh, Y. Huang, R. Jia, and J. Thomason. Does VLN pretraining work with nonsensical or irrelevant instructions? In *Open-Domain Reasoning Under Multi-Modal Settings (O-DRUM@CVPR)*, 2023.
- [19] **W. B. Zhu**, I. Singh, R. Jia, and J. Thomason. Language models can infer action semantics for symbolic planners from environment feedback. In *Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)*, 2025.
- [20] **W. B. Zhu**, J. Thomason, and R. Jia. Generalization differences between end-to-end and neuro-symbolic vision-language reasoning systems. In Y. Goldberg, Z. Kozareva, and Y. Zhang, editors, *Findings of EMNLP*, 2022.
- [21] **W. B. Zhu**, J. Thomason, and R. Jia. Chain-of-Questions training with latent answers for robust multistep question answering. In *Empirical Methods in Natural Language Processing (EMNLP)*, Dec. 2023.